

Population and Quantitative Trait Genetics

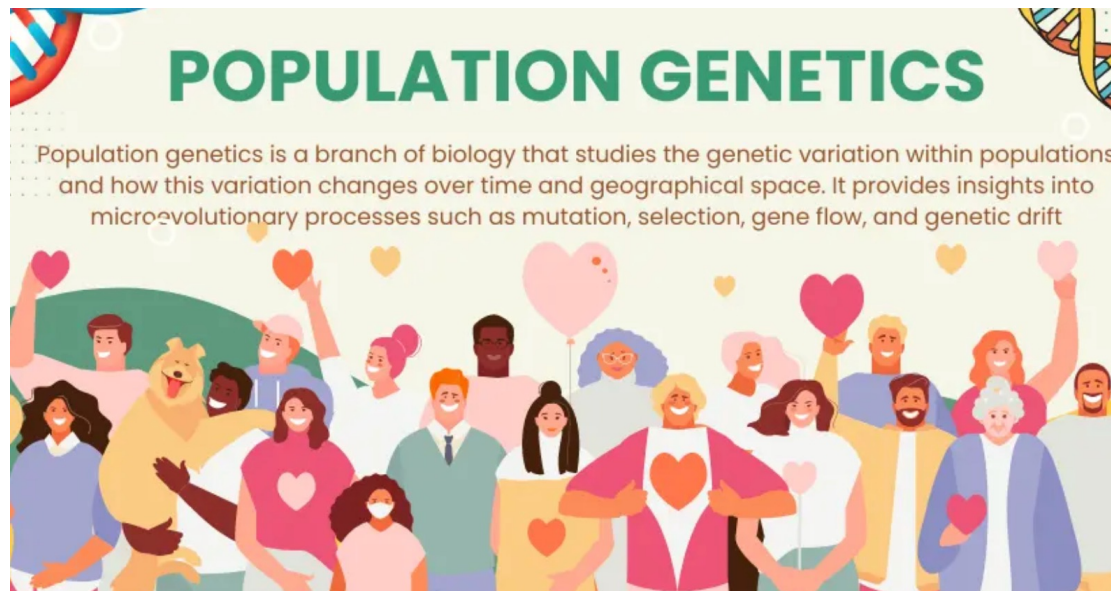
Prof. Schoen

Module: Population Genetics (first 2.5 lectures)

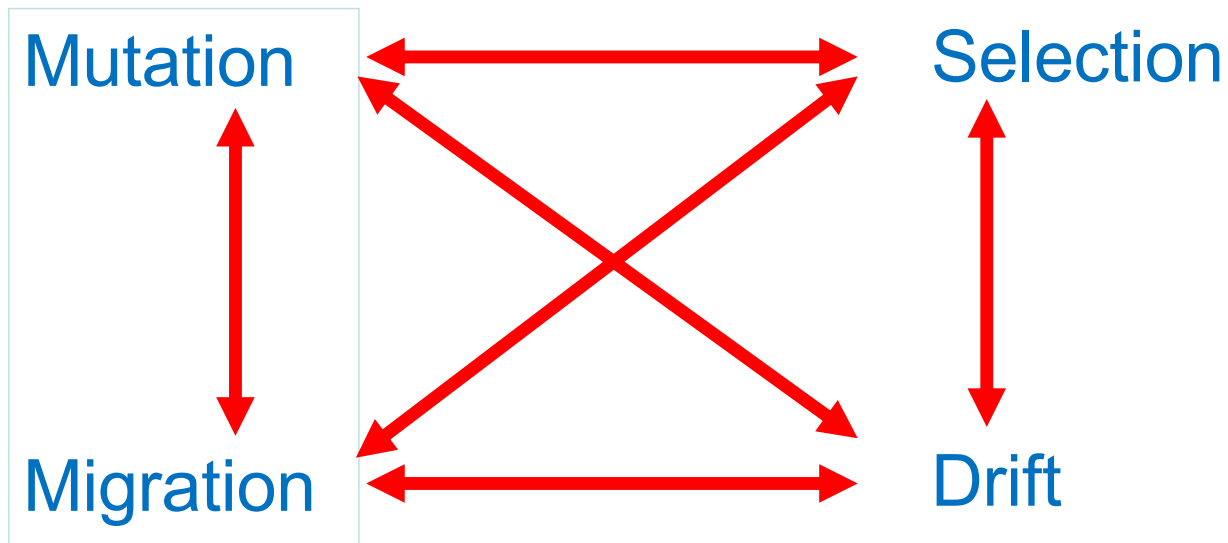
Lecture 10 — Genes in populations. Hardy-Weinberg theory.

Lecture 11 — Inbreeding. Mutation and migration. Genetic drift.

Lecture 12 — The genetics of natural selection. Intro to quantitative genetics.



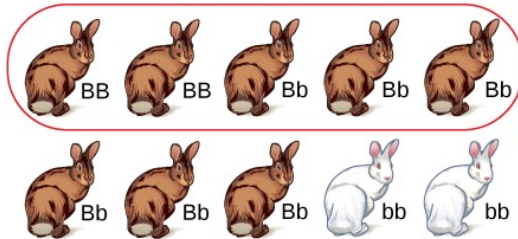
Evolutionary processes influence genotype and allele frequencies in populations



Genetic Drift

First generation

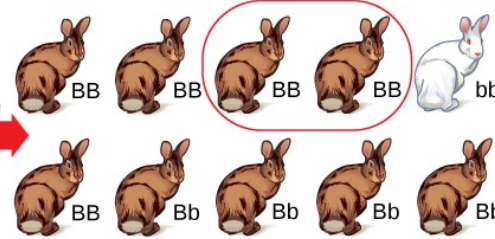
p (B gene frequency) = .5
 q (b gene frequency) = .5



5 rabbits
reproduce

Second generation

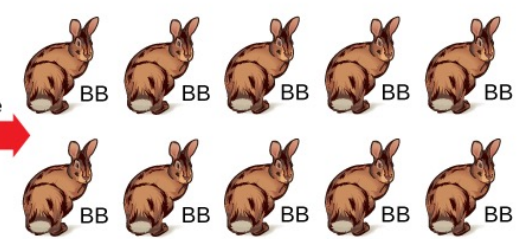
p = .7
 q = .3



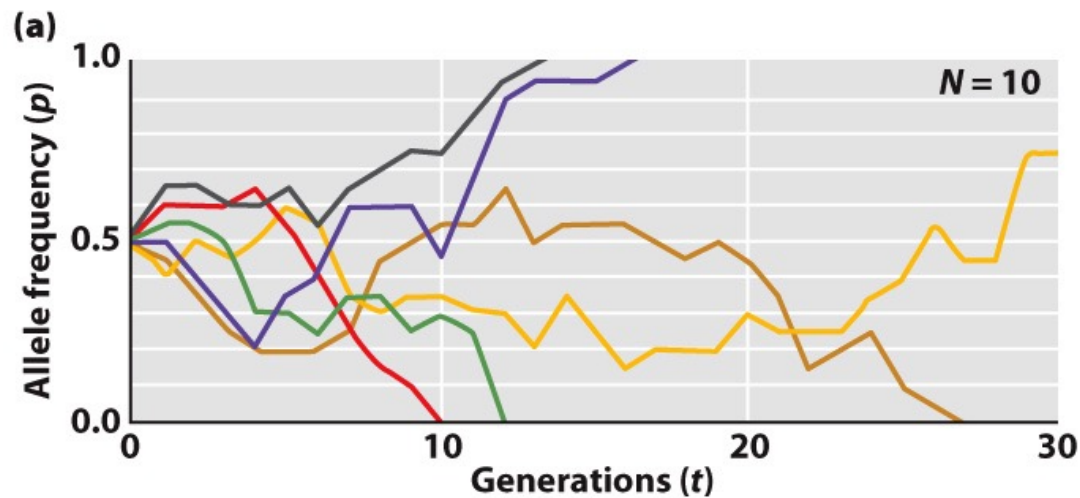
2 rabbits
reproduce

Third generation

p = 1
 q = 0



Random genetic drift is strong in small populations

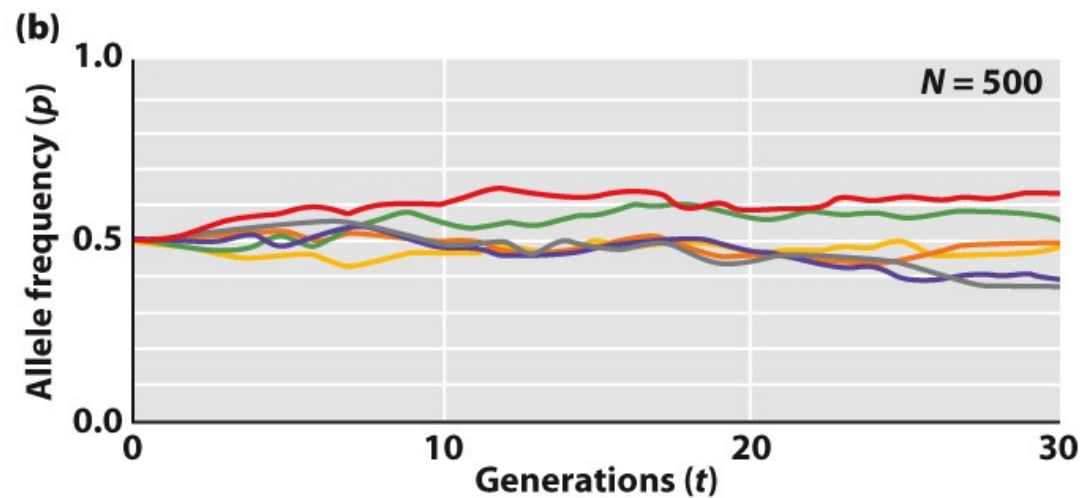


Each colour represents the frequency of one of the alleles at a biallelic locus

(b)

Allele frequency (p)

Random genetic drift is weaker in large populations



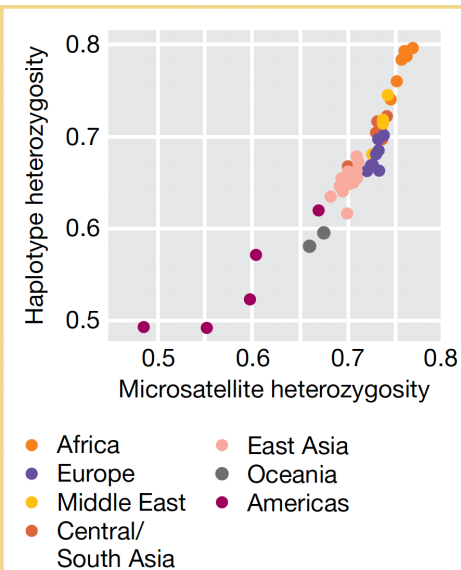
Each colour represents the frequency of one of the alleles at a biallelic locus

Genetic Drift (“founding events”)

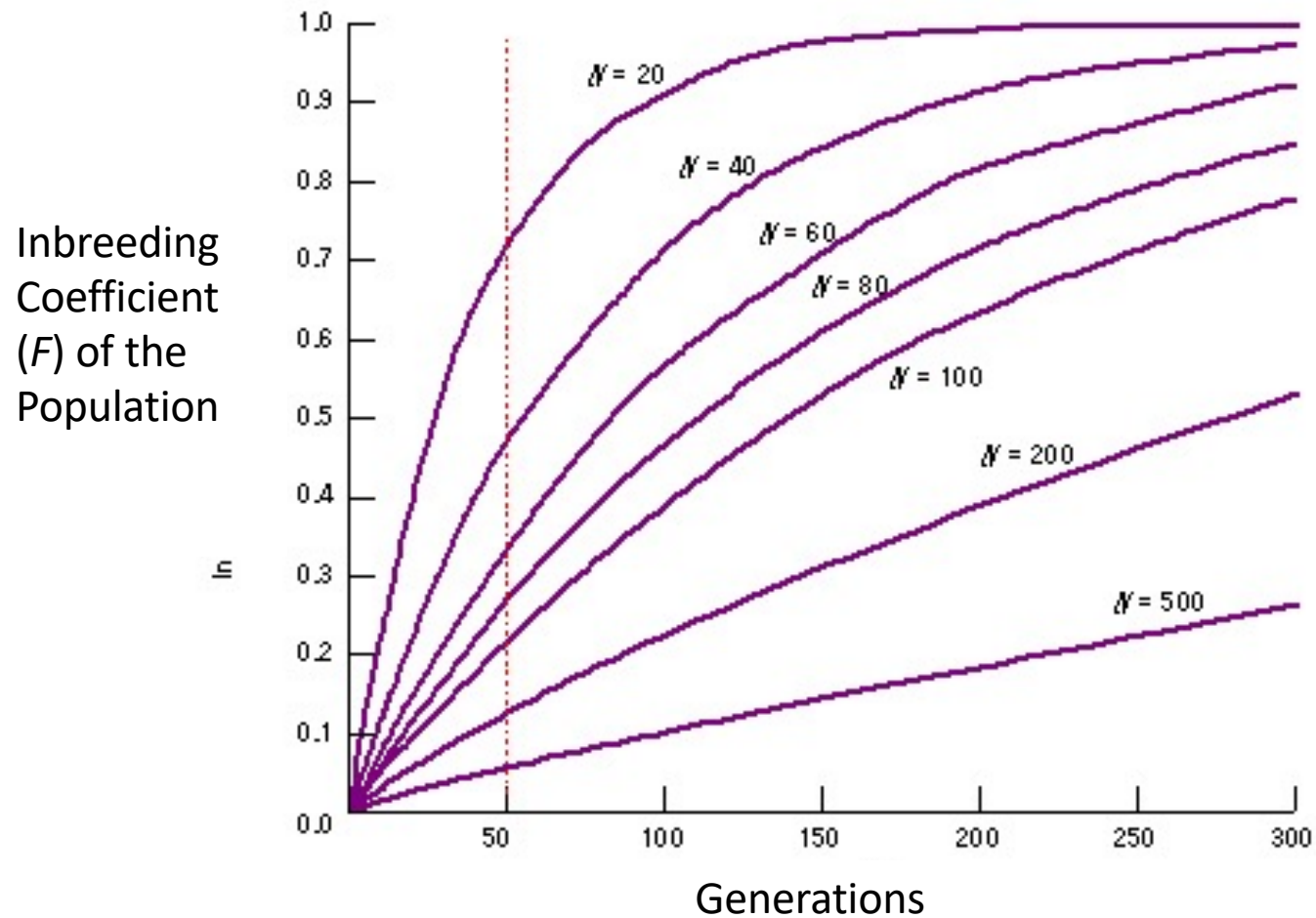
Figure 1: Human migration out of Africa.



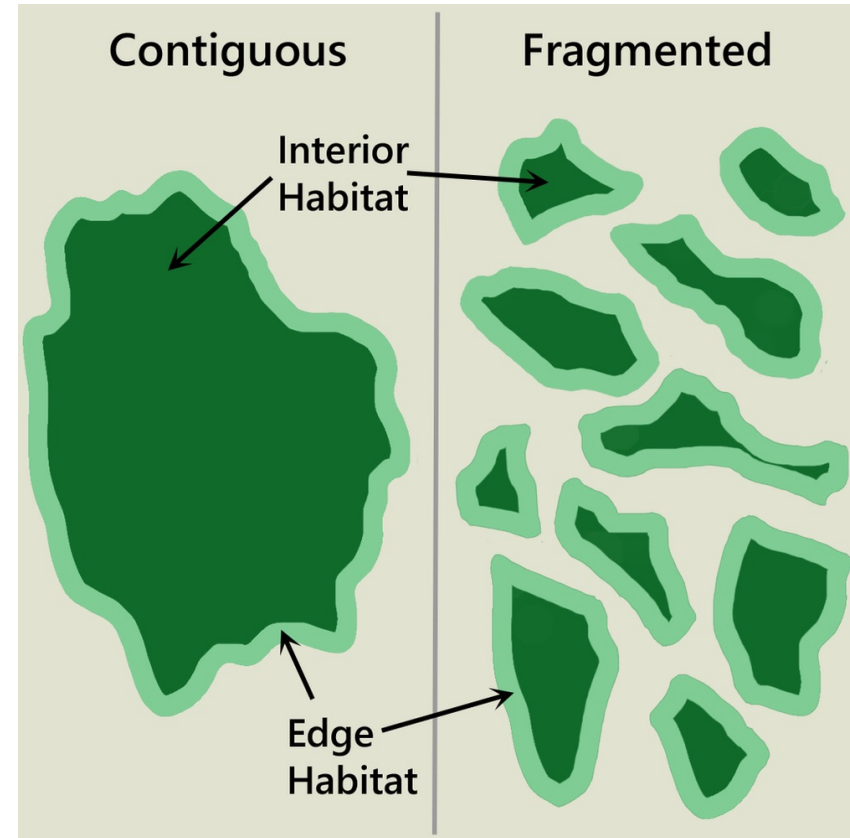
Human populations have different levels of genetic diversity



Successive generations of small population size → Inbreeding levels build up



Habitat fragmentation can reduce population sizes



Article

Global meta-analysis shows action is needed to halt genetic diversity loss

<https://doi.org/10.1038/s41586-024-08458-x>

Received: 11 December 2023

Accepted: 27 November 2024

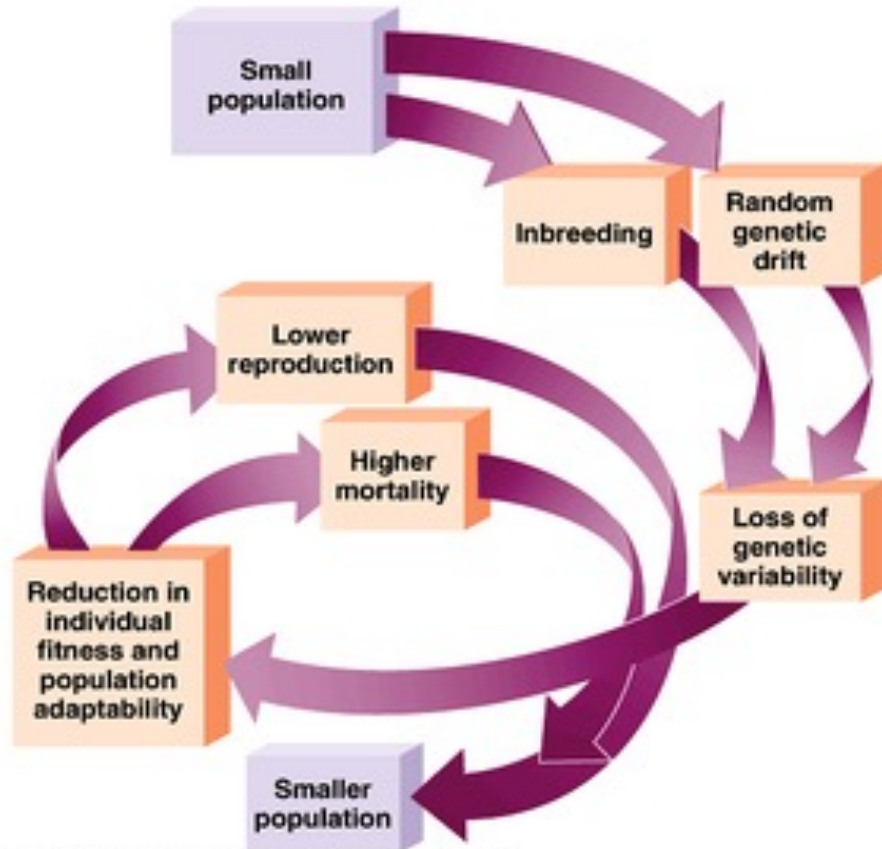
Published online: 29 January 2025

Open access

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Small populations and the “Extinction Vortex”

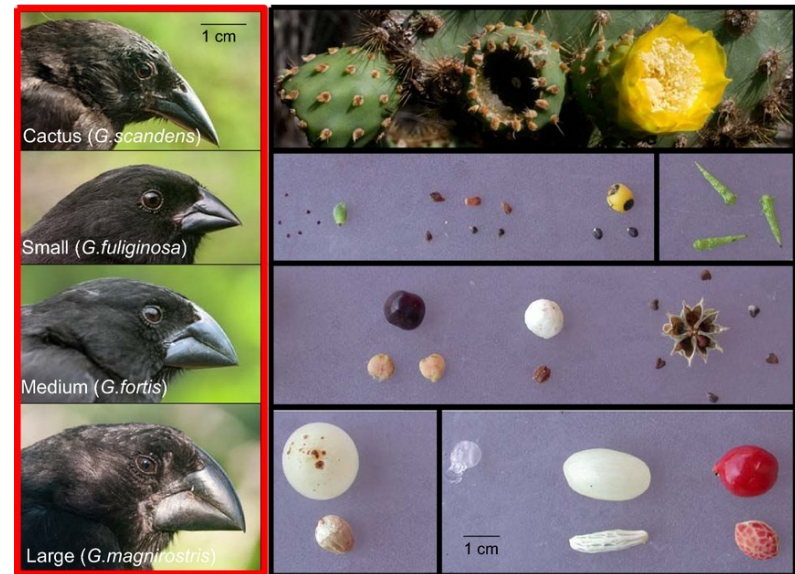


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Processes influencing allele frequencies

- *Mutation*
- *Migration*
- *Genetic Drift*
- *Natural Selection*



Module: Population Genetics



Lecture 12—The population genetics of natural selection

- Genetics of Natural Selection (Ch18 – section 5) skip subsection on "Recombination" and "Linkage Disequilibrium" and Box 18.5)
- Quantitative (or complex) trait genetics: genes and environment (Ch19 section 1)

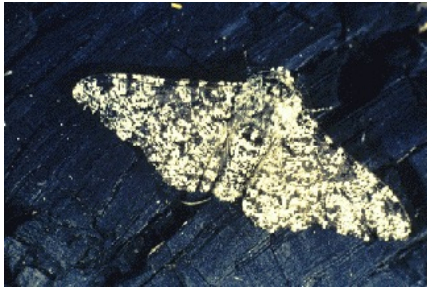
Population genetic models of selection

allow us to:

- Predict the final outcome of selection (i.e., the equilibrium allele frequency at a locus)
- Predict rate of evolutionary change (i.e., the rate of change in allele frequency at a locus)
- Examine the interaction of selection with other population processes.

Natural selection in mid-19th Britain favoured the dark form of the Peppered Moth

*Light
Form*



*Dark
Form*



*Hungry
sparrow*



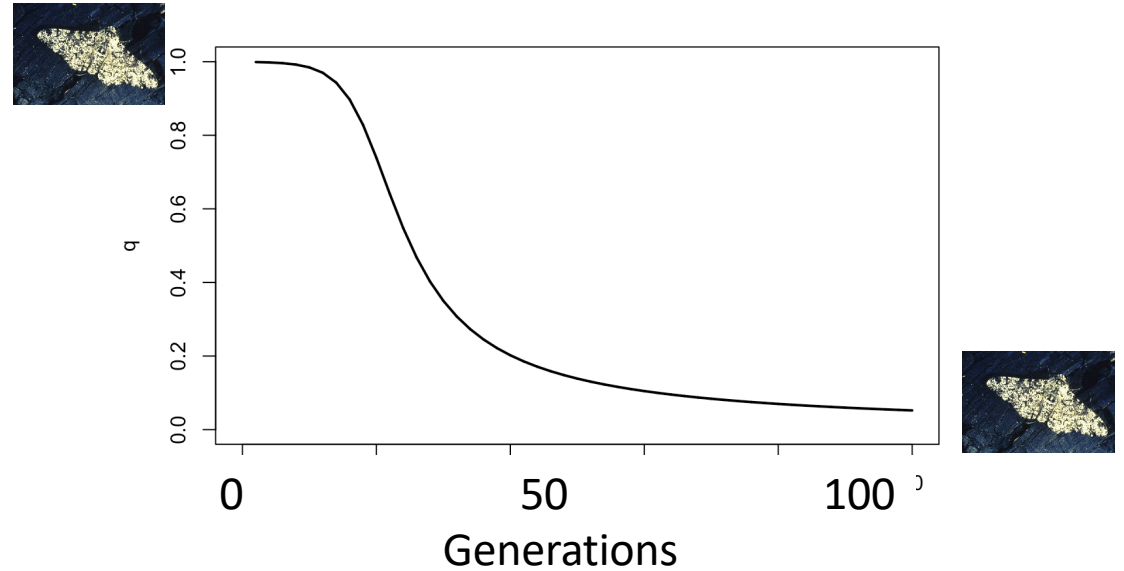
*When coal was “king”
(mid-19th century Britain)*

During the era of heavy coal burning in Britain, it took about a hundred years for the white form of the Peppered Moth to be nearly replaced by the dark form

*Light
Form*



*Dark
Form*



“light form”

“dark form”

Genotype aa

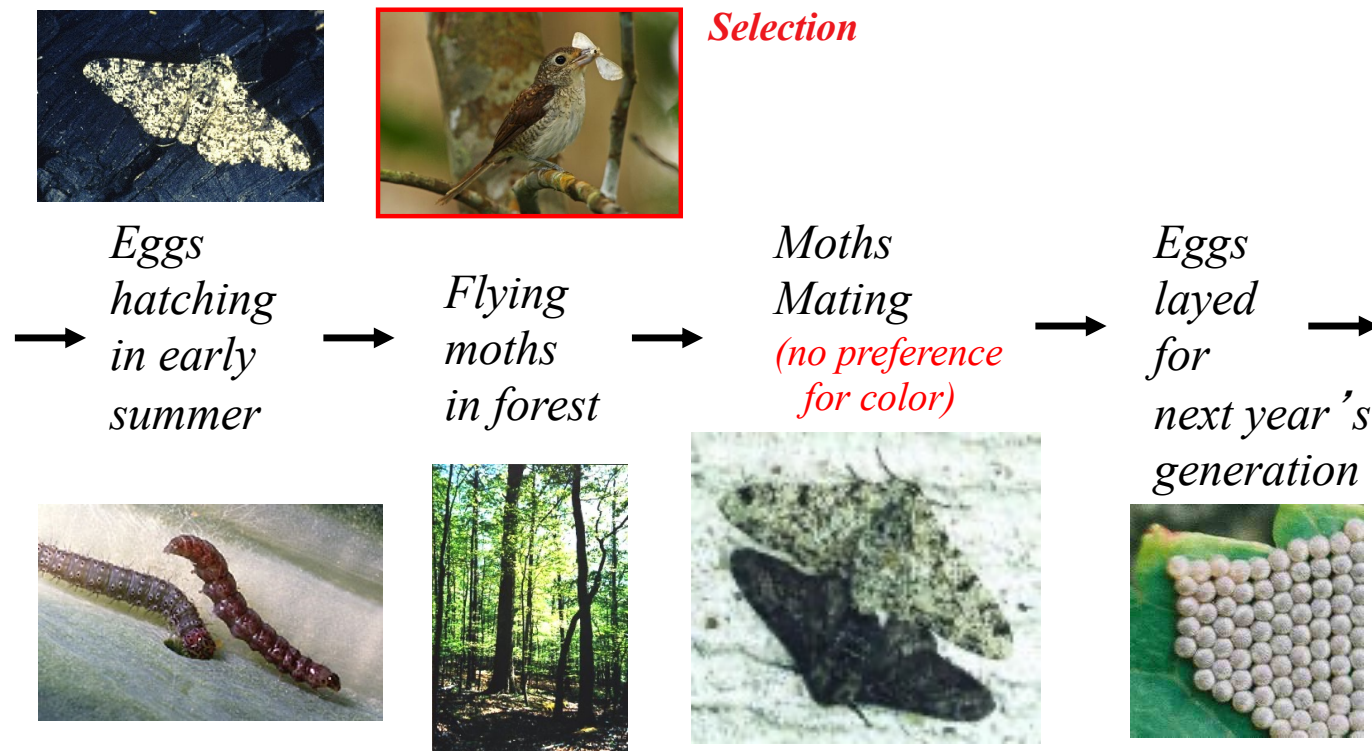


Genotype AA or Aa



Peppered Moth

How will natural selection act to alter frequency of allele **a** (“light”) in a coal-polluted forest environment?



Natural selection typically works by favouring the form with the highest **relative fitness**

Genotype	Absolute Fitness <i>(Percent survival to reproductive age)</i>	Relative Fitness <i>(in coal-polluted forest)</i> <i>where s = selection coefficient</i>
<i>AA (dark)</i>	30%	$w_{AA} = 1.0$
<i>Aa (dark)</i>	30%	$w_{Aa} = 1.0$
<i>aa (light)</i>	21%	$w_{aa} = 0.7$ (i.e., $s = 0.3$)

Using a model of selection to calculate genotype frequencies after selection

<u>Genotype</u>	<u>Phenotype</u>	<u>Relative fitness</u>	<u>Starting frequencies the genotypes</u>	<u>Genotype frequencies weighted by fitness</u>
AA	dark	$w_{AA} = 1$	$f_{AA} = 0.01$	$w_{AA} f_{AA} = 0.01$
Aa	dark	$w_{Aa} = 1$	$f_{Aa} = 0.18$	$w_{Aa} f_{Aa} = 0.18$
aa	light	$w_{aa} = 1 - s$ $= 0.7$	$f_{aa} = 0.81$	$w_{aa} f_{aa} = 0.567$

$$\begin{aligned}
 W_{avg} &= f_{AA} w_{AA} + f_{Aa} w_{Aa} + f_{aa} w_{aa} \\
 &= 0.01 (1) + 0.18 (1) + 0.81 (0.7) \\
 &= \mathbf{0.757}
 \end{aligned}$$

NOTE: A population average value is equal to the sum of individual values times their respective frequencies

Using a model of selection to calculate genotype frequencies after selection

<u>Genotype</u>	<u>Phenotype</u>	<u>Relative fitness</u>	<u>Starting frequencies the genotypes</u>	<u>Genotype frequencies weighted by fitness</u>	<u>Genotype frequencies after selection</u>
AA	dark	$w_{AA} = 1$	$f_{AA} = 0.01$	$w_{AA} f_{AA} = 0.01$	$w_{AA} f_{AA} / W_{avg} = 0.0132$
Aa	dark	$w_{Aa} = 1$	$f_{Aa} = 0.18$	$w_{Aa} f_{Aa} = 0.18$	$w_{Aa} f_{Aa} / W_{avg} = 0.2378$
aa	light	$w_{aa} = 1-s$ $= 0.7$	$f_{aa} = 0.81$	$w_{aa} f_{aa} = 0.567$	$w_{aa} f_{aa} / W_{avg} = 0.7490$

The frequency of the "a" allele **after selection** = $0.2378/2 + 0.7490 = 0.8679$

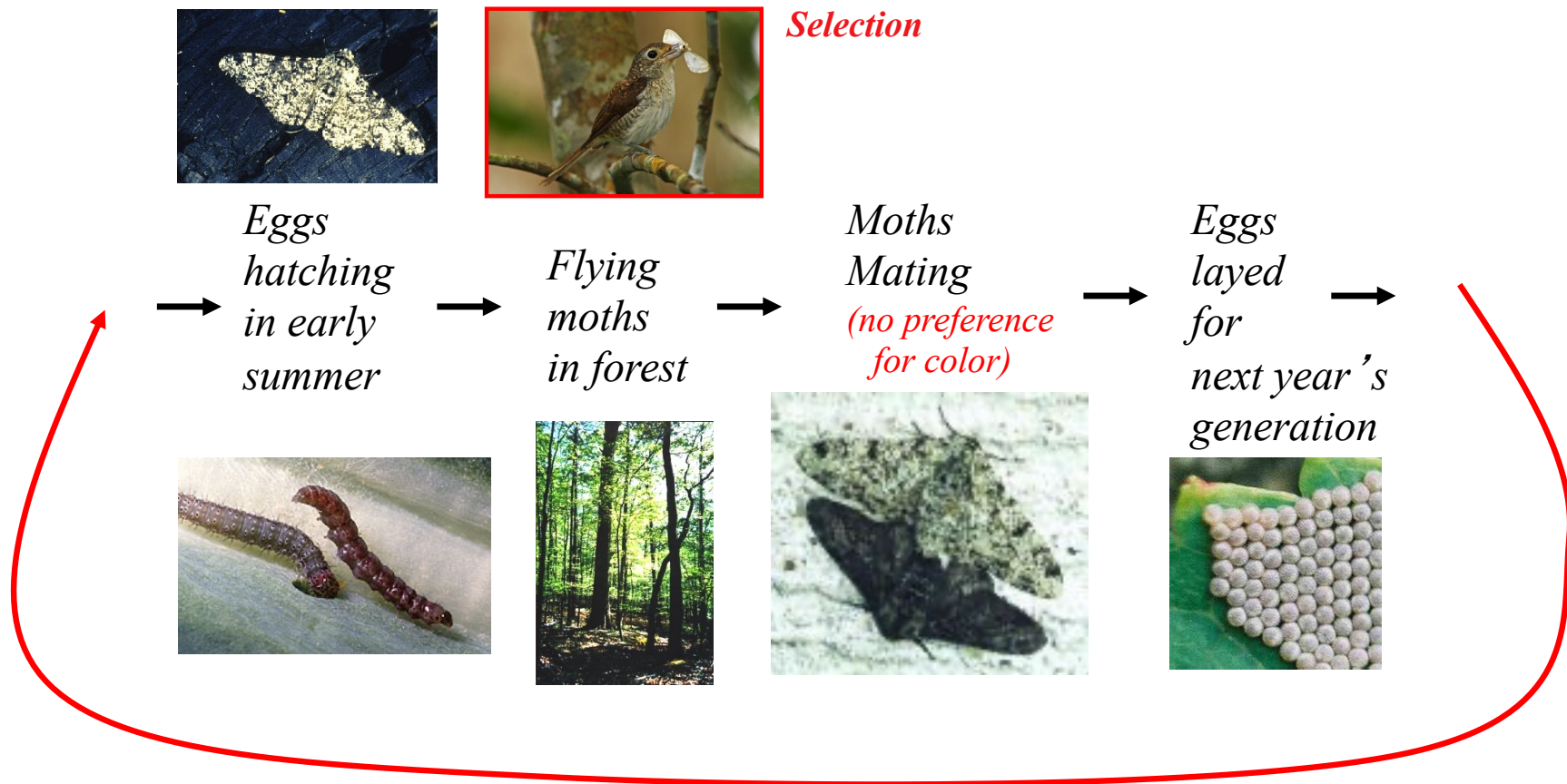
Using a model of selection to calculate genotype frequencies after selection

<u>Genotype</u>	<u>Phenotype</u>	<u>Relative fitness</u>	<u>Starting frequencies the genotypes</u>	<u>Genotype frequencies weighted by fitness</u>	<u>Genotype frequencies after selection</u>
AA	dark	$w_{AA} = 1$	$f_{AA} = 0.01$	$w_{AA} f_{AA} = 0.01$	$w_{AA} f_{AA} / W_{avg} = 0.0132$
Aa	dark	$w_{Aa} = 1$	$f_{Aa} = 0.18$	$w_{Aa} f_{Aa} = 0.18$	$w_{Aa} f_{Aa} / W_{avg} = 0.2378$
aa	light	$w_{aa} = 1-s$ $= 0.7$	$f_{aa} = 0.81$	$w_{aa} f_{aa} = 0.567$	$w_{aa} f_{aa} / W_{avg} = 0.7490$

The frequency of the “a” allele **after selection** = $0.2378/2 + 0.7490 = 0.8679$

After one generation of selection, the frequency of the “a” allele went from **0.9 → 0.8679**

How will natural selection act to alter frequency of allele **a** (“light”) in a coal-polluted forest environment?



Directional selection will eventually drive the less fit form from the population

Natural selection acting on a gene involved in pigmentation

"light form"

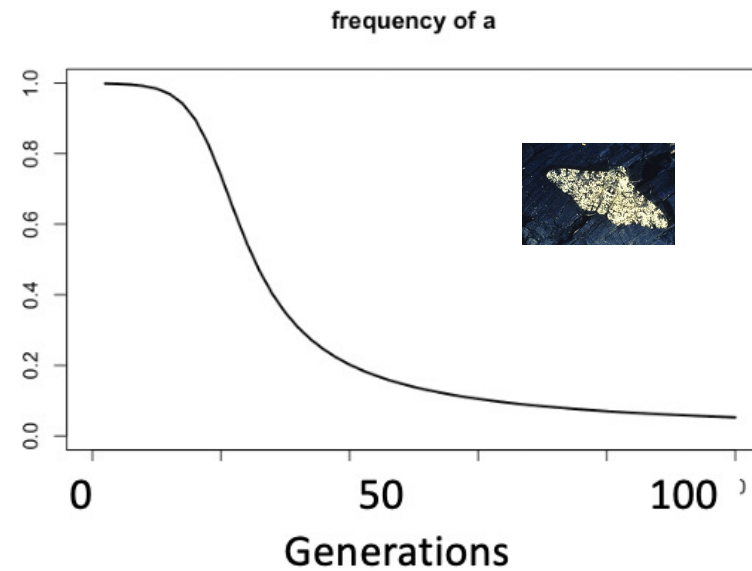
"dark form"

Genotype aa

Genotype AA or Aa

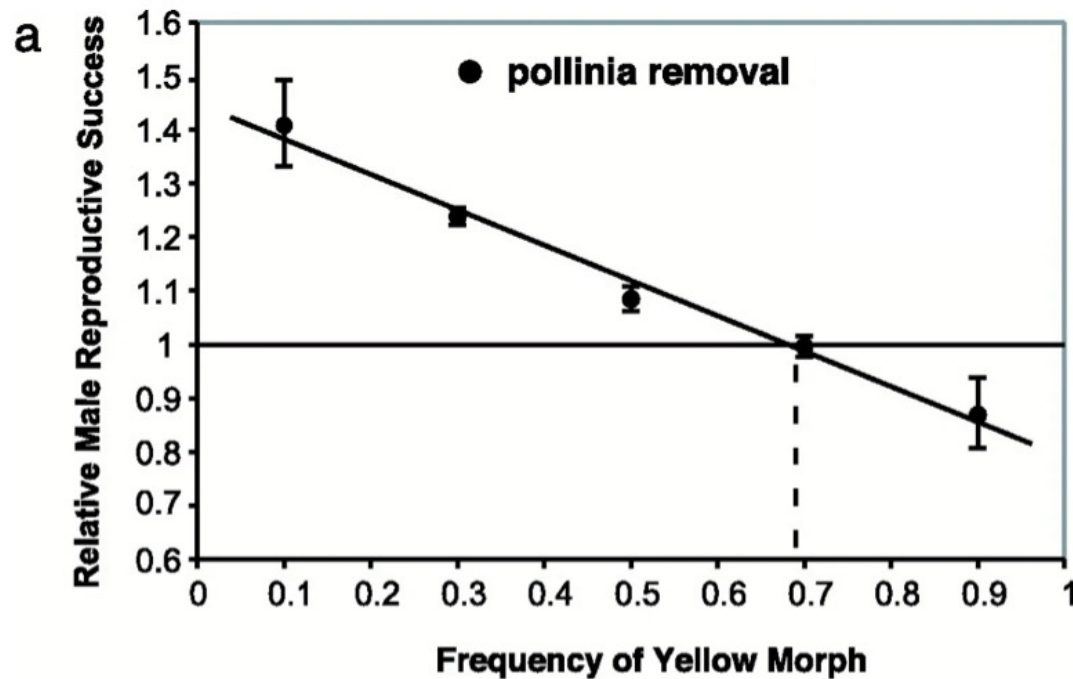


Peppered Moth



We've just considered an example of **DIRECTIONAL** selection,
but **BALANCING** selection can also occur in nature

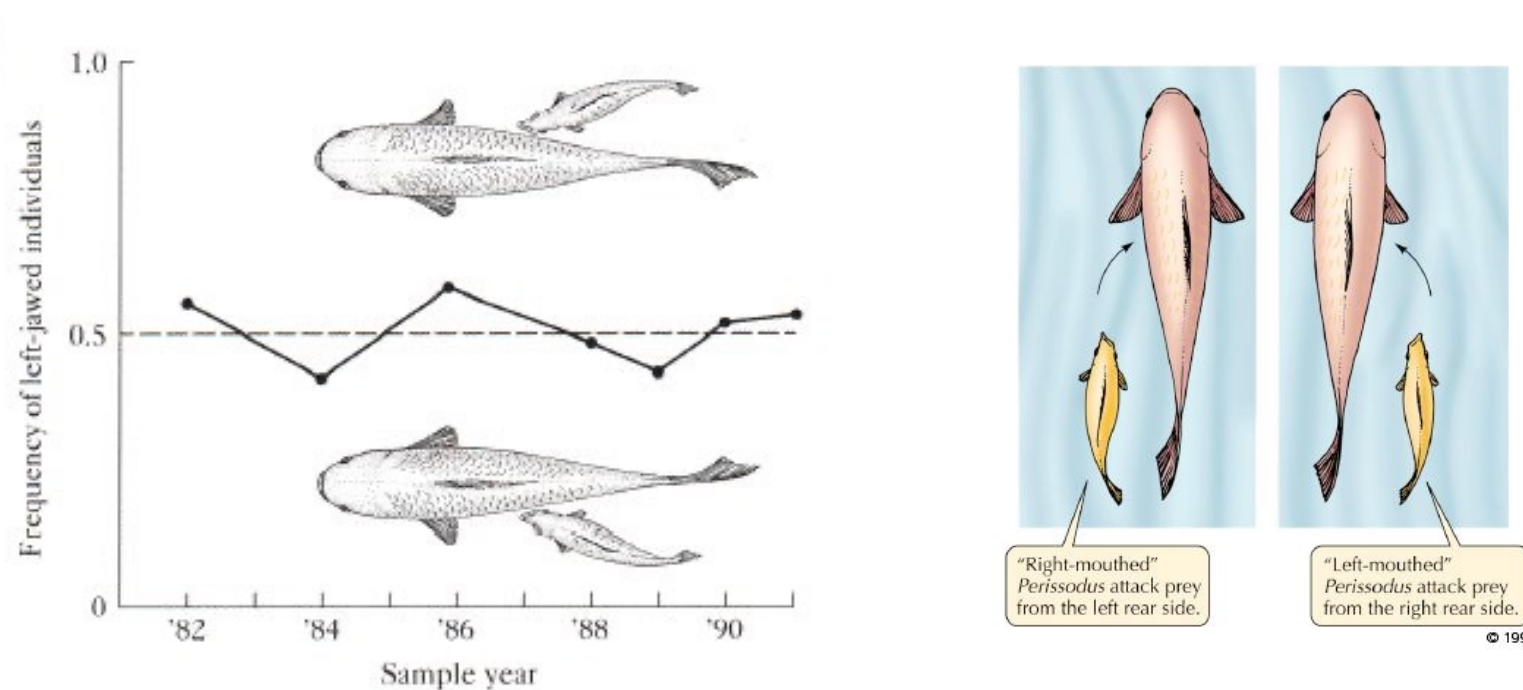
(frequency-dependent selection is a type of balancing selection)



*Frequency-
dependent
selection of
flower color*

We've just considered an example of **DIRECTIONAL** selection,
but **BALANCING** selection can also occur in nature

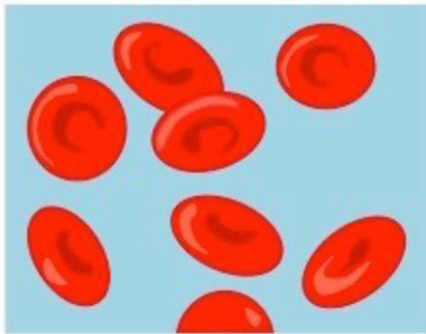
(frequency-dependent selection is a type of balancing selection)



Predator-prey relations in Lake Tanganyika cichlid fish that eat other fishes' scales.

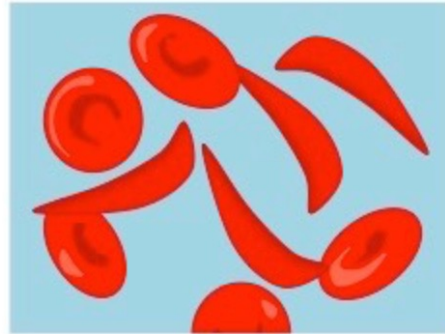
We've just considered an example of **DIRECTIONAL** selection,
but **BALANCING** selection can also occur in nature

(Heterozygote advantage is another type of balancing selection)



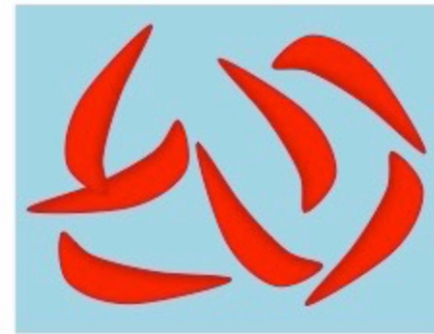
AA

Susceptible to malaria
but no sickle cell disease



Aa

Resistant to malaria
and only mild sickle cell disease



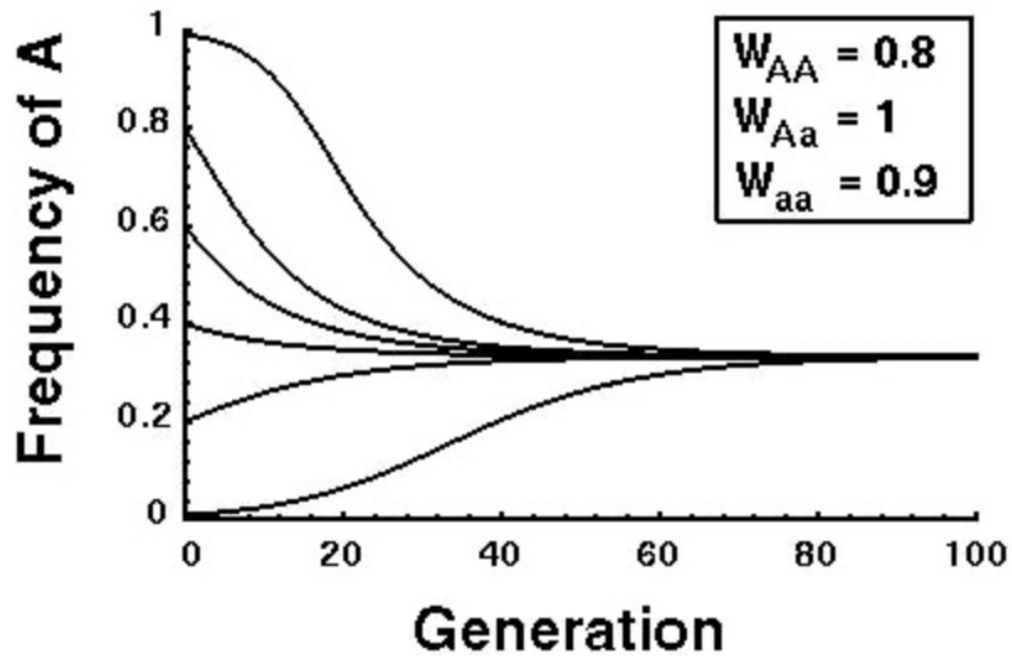
aa

Resistant to malaria
but has fatal sickle cell disease

Model for heterozygote advantage

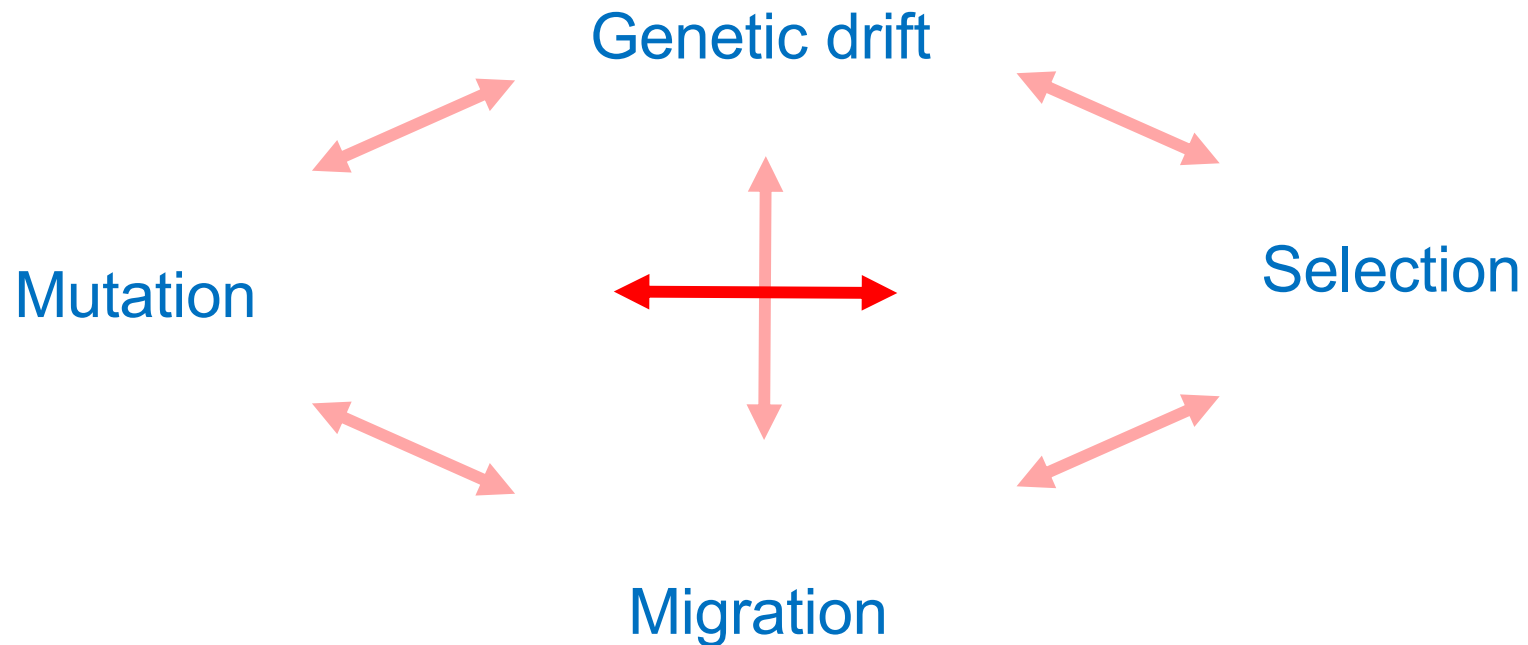
<u>Genotype</u>	<u>Form</u>	<u>Relative fitness</u>
<i>AA</i>	<i>phenotype 1</i>	$w_{AA} = 1-t$
<i>Aa</i>	<i>phenotype 2</i>	$w_{Aa} = 1$
<i>aa</i>	<i>phenotype 3</i>	$w_{aa} = 1-s$

Allele frequency change under heterozygote advantage

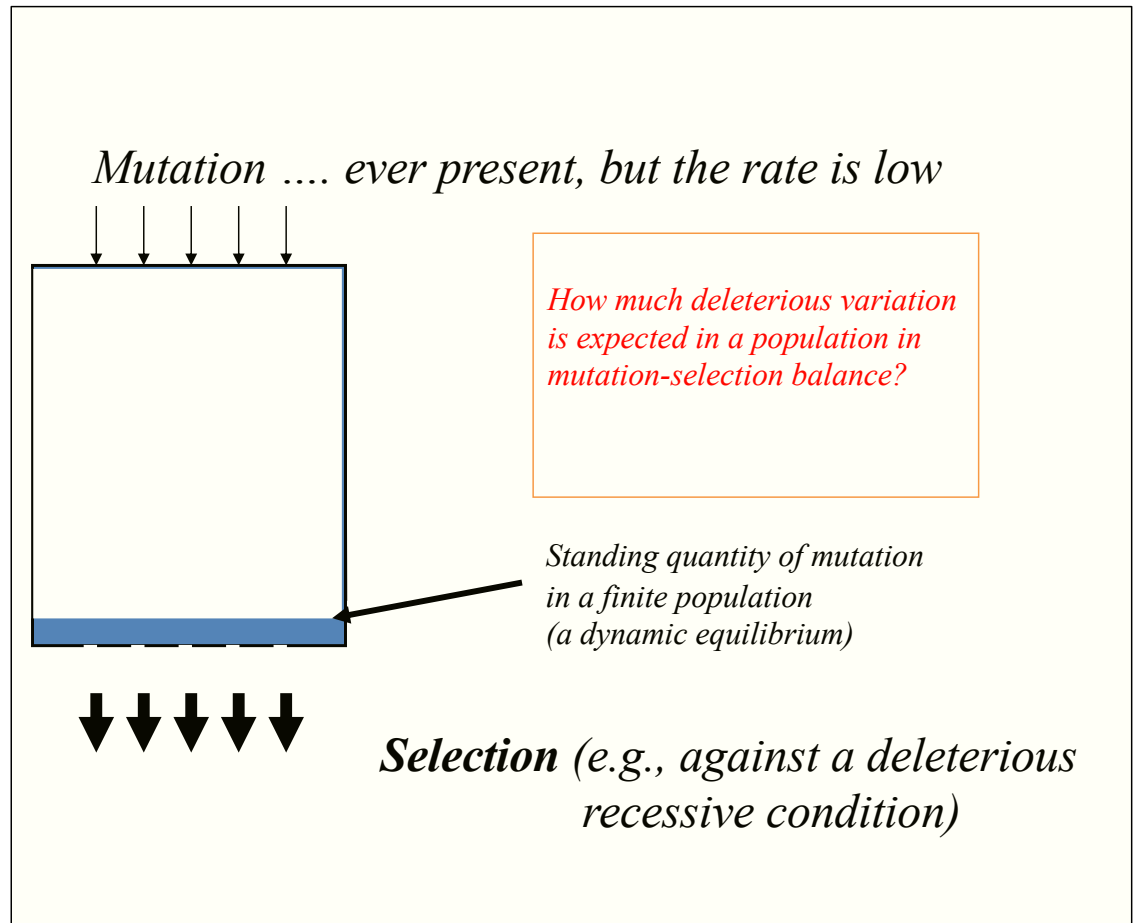
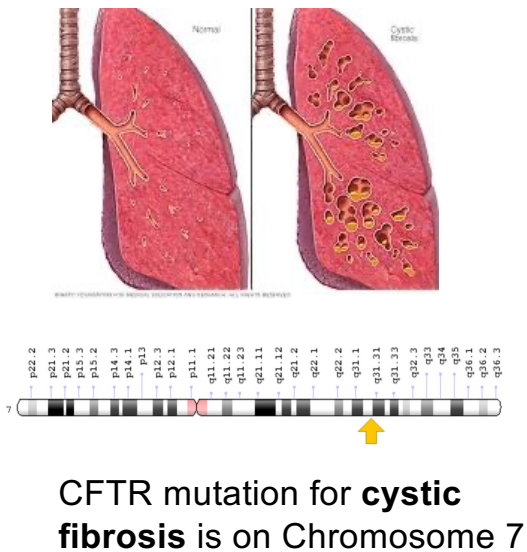


*Both A and a alleles
are maintained*

Processes influencing allele frequencies **interact !**



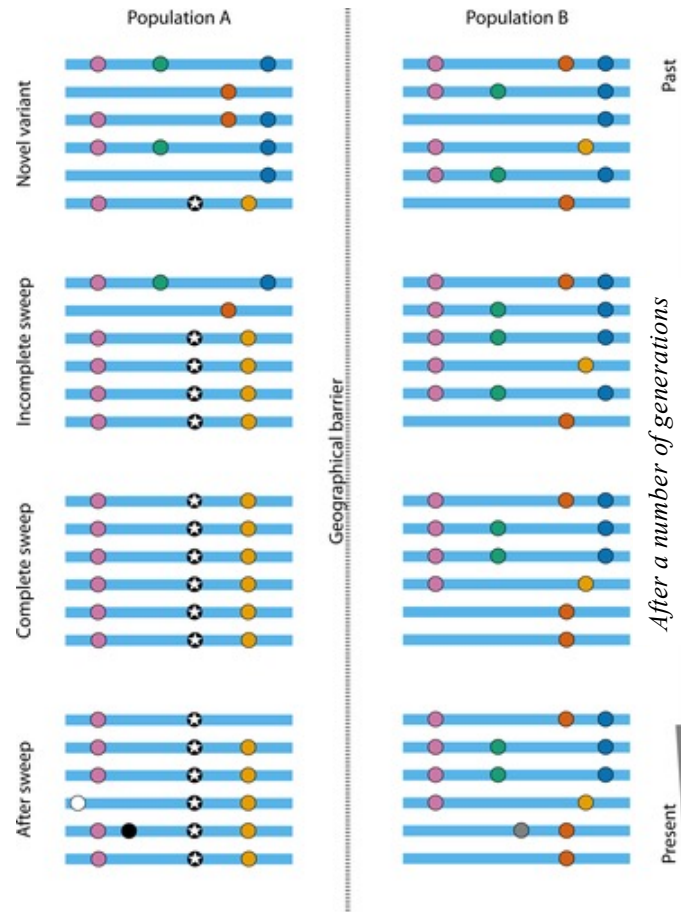
Deleterious mutation-selection interaction (why bad things happen to good populations)



The B locus has two alleles B and b with frequencies of $p = 0.95$ and $q = 0.05$, respectively. The genotypic relative fitnesses at this locus are $w_{B/B} = 0.8$, $w_{B/b} = 1.0$, and $w_{b/b} = 0.7$. What will the frequency of the b allele be in the next generation?

- A. 0.0602
- B. 0.1182
- C. 0.0501
- D. 0.1002

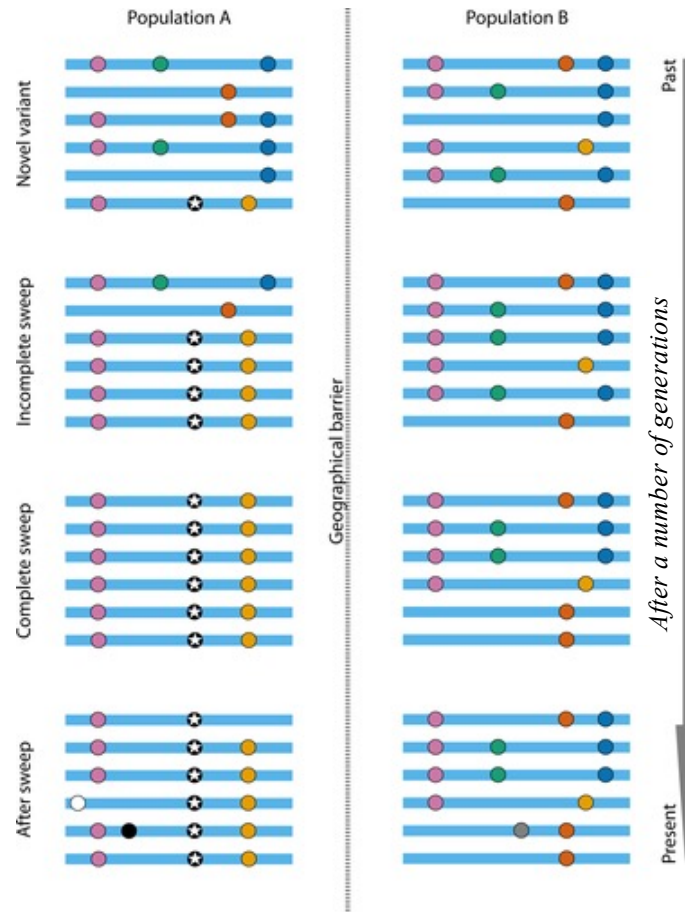
Selective sweeps can reduce diversity in genomic regions, especially those with low rates of recombination—they leave behind a trace of past events that have influenced the genome.



We have detected “GHOST OF SELECTION PAST” in Population A.



Selective sweeps can reduce diversity in genomic regions, especially those with low rates of recombination



Note that this one chromosome becomes common as a result of the sweep in the Population (Population A) where the ★ nucleotide mutation is strongly selected

(In Population B the ★ nucleotide mutation did not occur)

Lactase allele variant that causes lactase to persist through lie

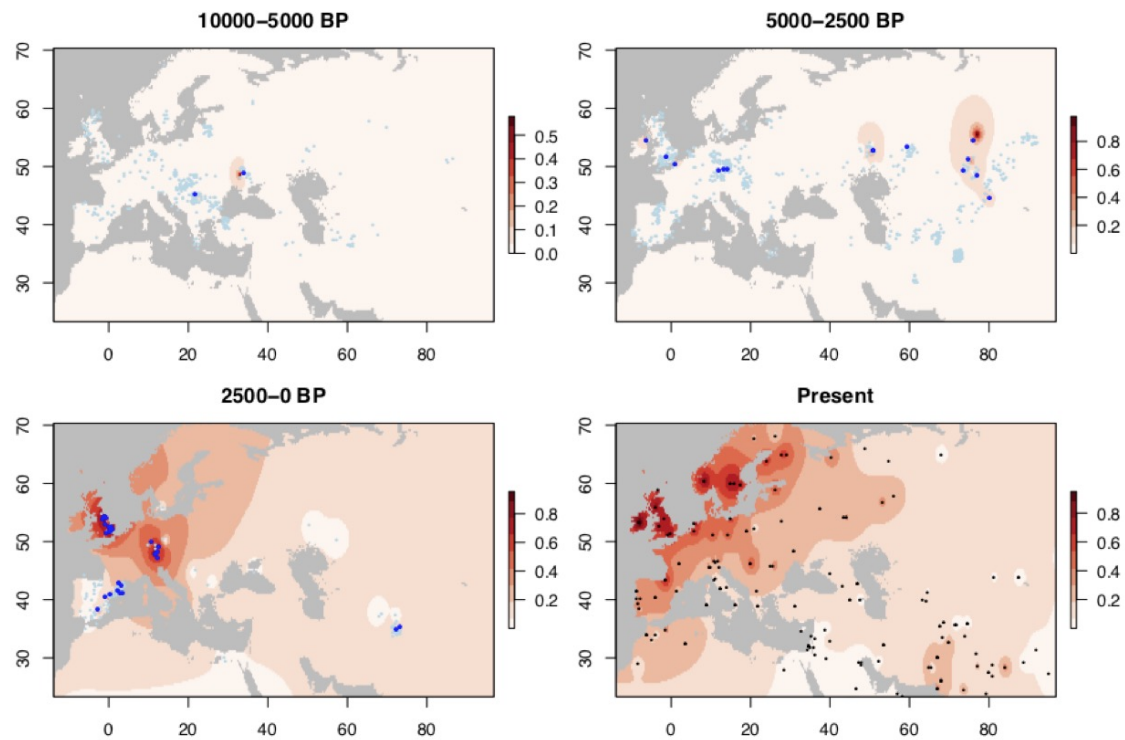
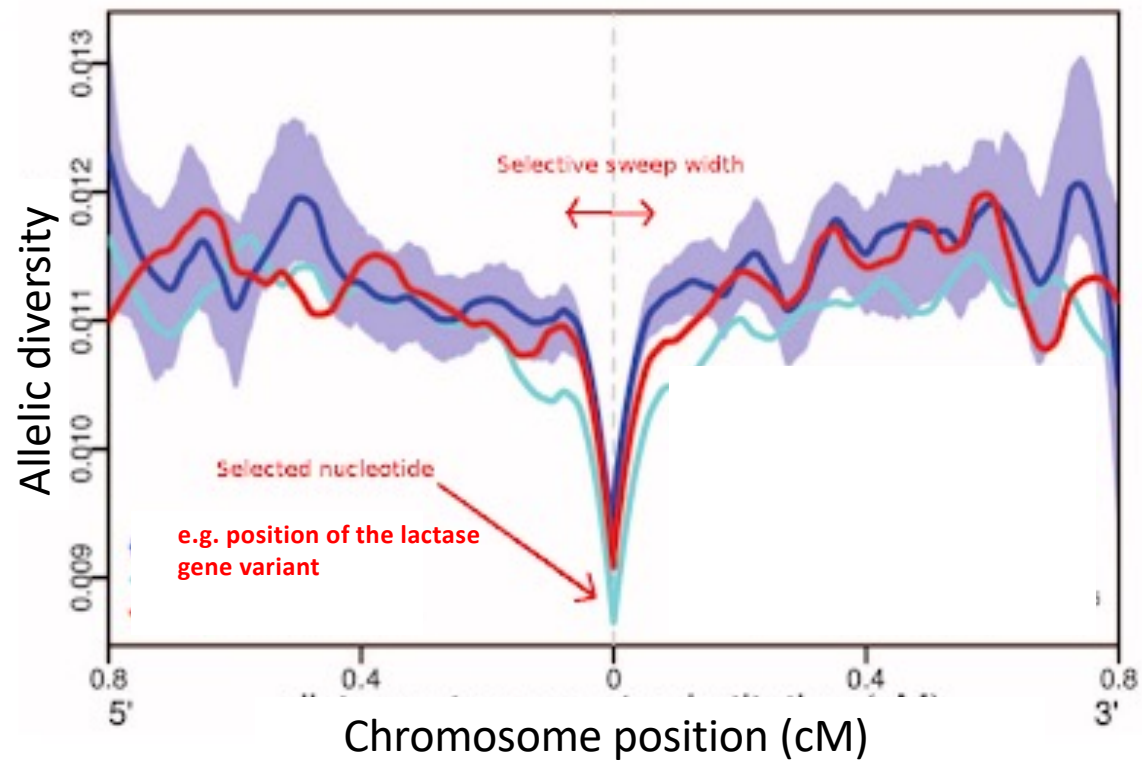


Figure 2.99: **Increase in allele frequency of the lactase variant in Europe (from ancient DNA).** The maps indicate locations of samples with ancestral alleles (light blue) and derived alleles (dark blue) in each time period (BP = years Before Present); as well as spatially smoothed maps of the derived allele frequency in each period (red color scale beside each panel). Credit: Iain Mathieson blog post (2019) [[Link](#)]. Used with permission of the author.

Selective sweep

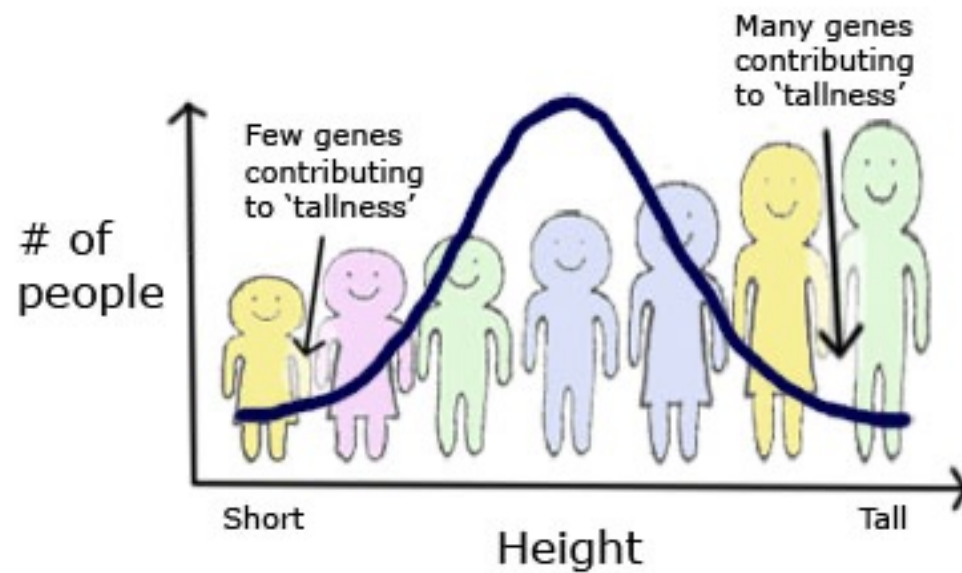


Main points—natural selection

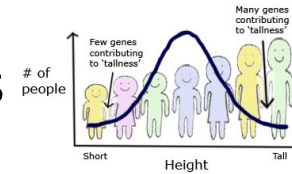
- Simple genetic models of selection predict allele frequency change over time
- Selection may be directional or balancing
- Genome level analyses can be used to detect past selection
- Selection and deleterious mutation when they interact can maintain genetic diseases in populations

Questions about population genetics?

Module: Complex trait genetics



Module – Complex Trait (Quantitative) Genetics



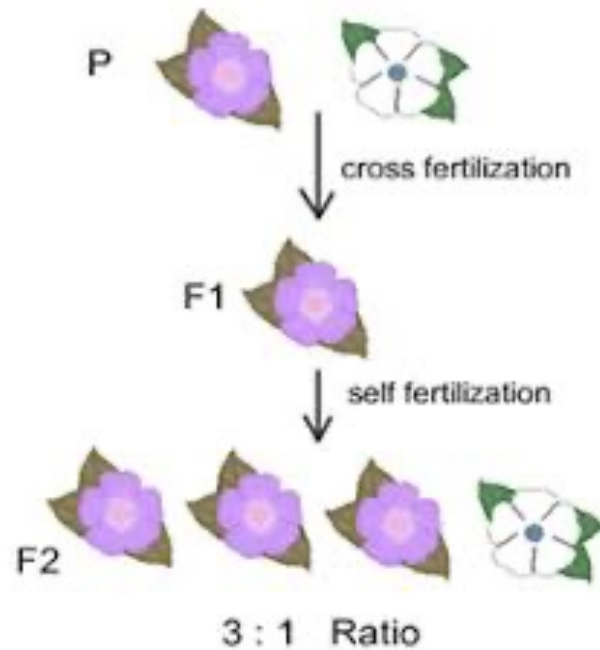
Lecture 12/13— When many genes and many environmental effects influence trait expression and variation

1. Complex trait variation (Ch19 Section 1)
2. Measuring components of phenotypic trait variation (Ch19 Sections 2,3)
3. Using molecular markers to infer the presence of genes affecting genetic variance (Ch19 Section 6)

PRIMER—Quantitative Genetics (see myCourses page for Biology 202)

Simple traits versus complex traits

Simple trait



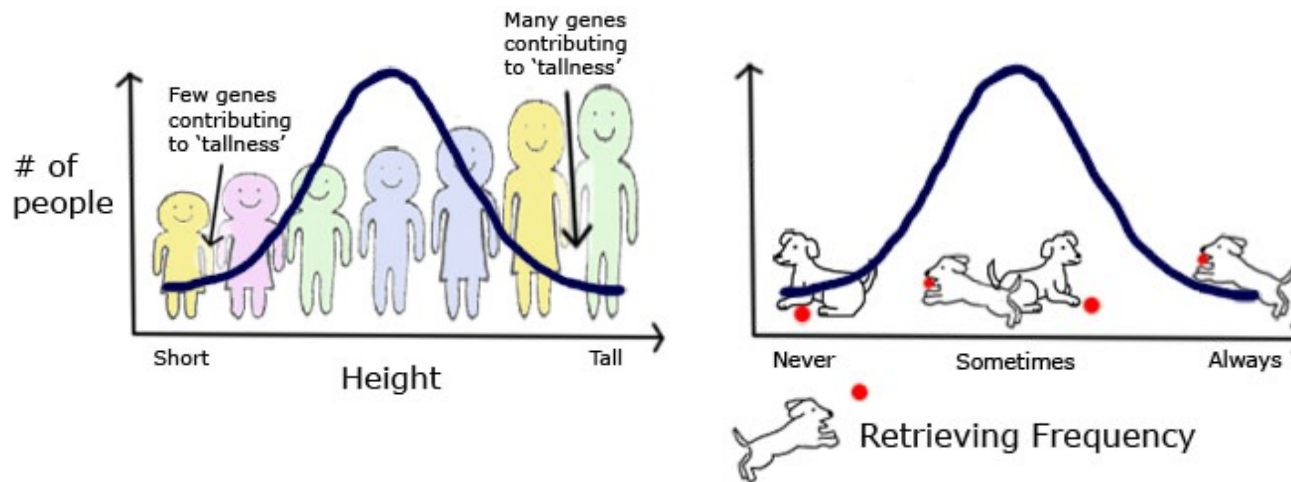
Complex (also called “quantitative” trait



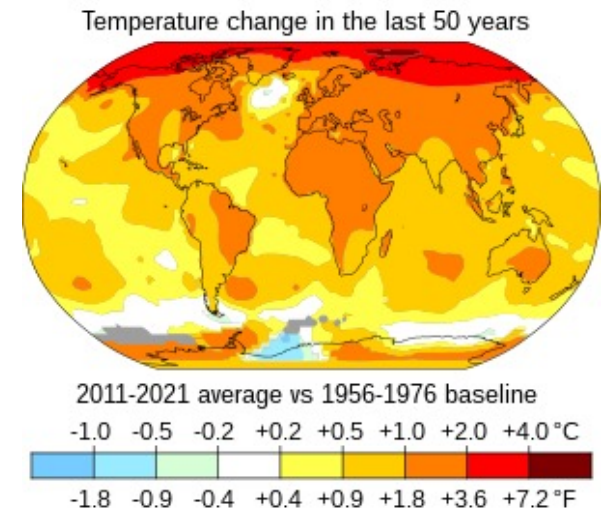
Population variation controlled by one /two genes by many (e.g., 100 genes)

Complex (quantitative) trait genetics

The nature of quantitative variation:
Distribution of phenotypic states is often “bell-shaped”



Quantitative genetic variation: Important for understanding the evolution of common diseases, crop improvement, and the rate of evolutionary change

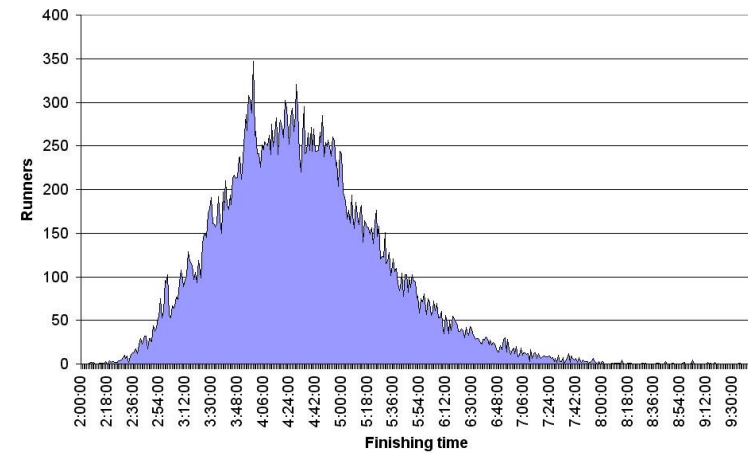


Types of complex traits

Continuous



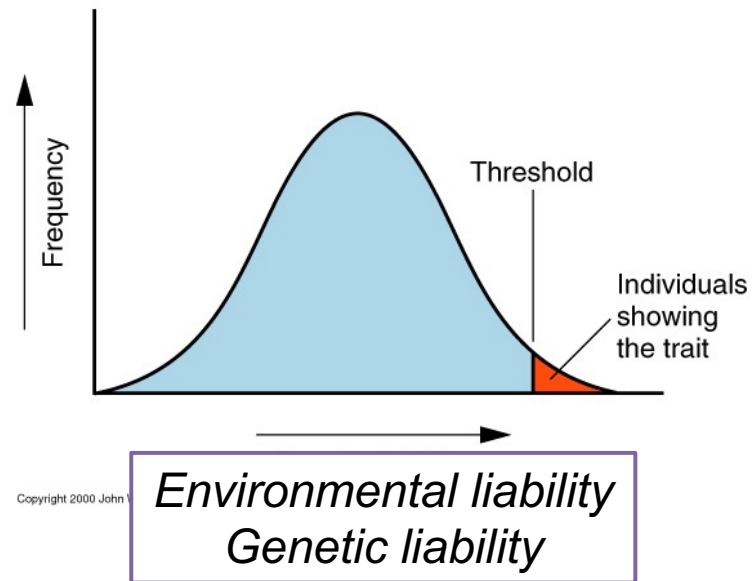
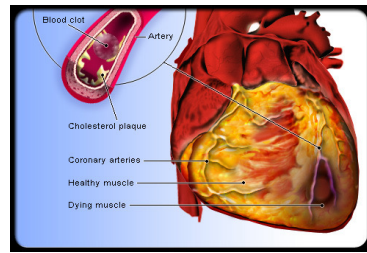
Body size (height)



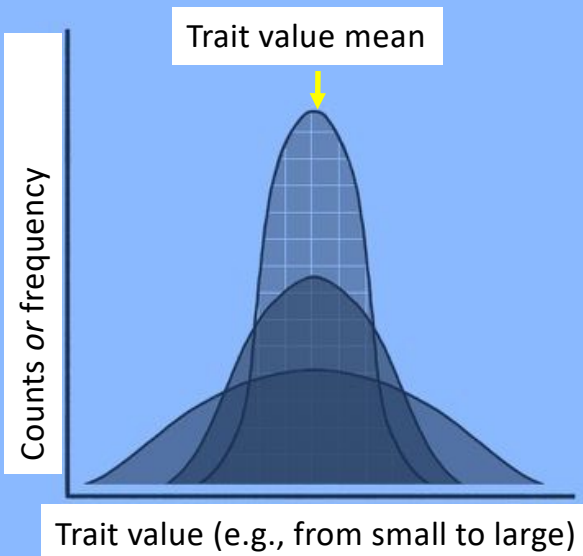
*Running speed
(e.g., finishing time in a marathon)*

Types of complex traits

Threshold (“either / or”)



Threshold traits
(e.g., cancers, heart diseases)



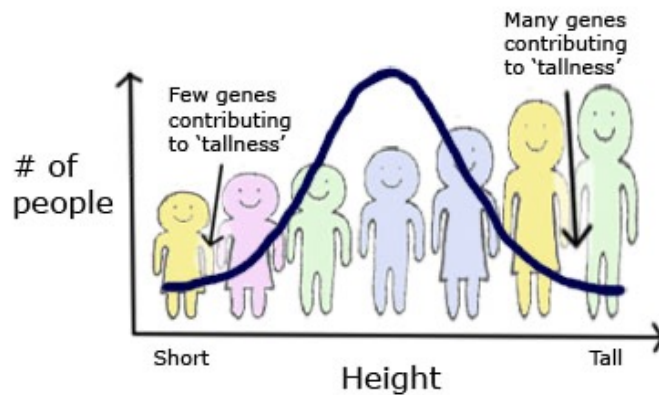
Variance

[ˈver-ē-ən(t)s]

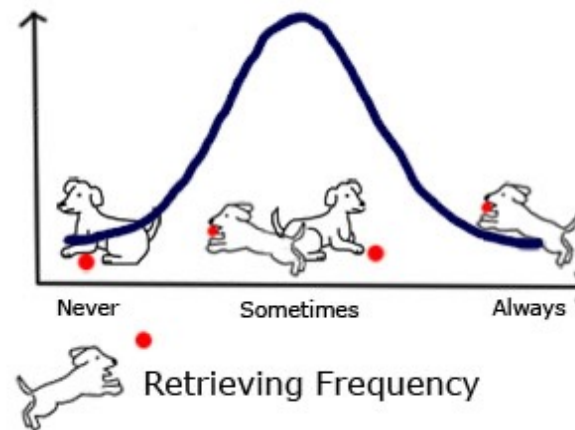
A measurement of how far each number in a data set is from the mean (average), and thus from every other number in the set.

Phenotypic variance of a trait: How much is due to differences in the environment that individuals experience *and* how much is due to genetic differences among individuals

Phenotypic variance of height



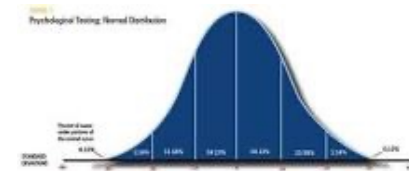
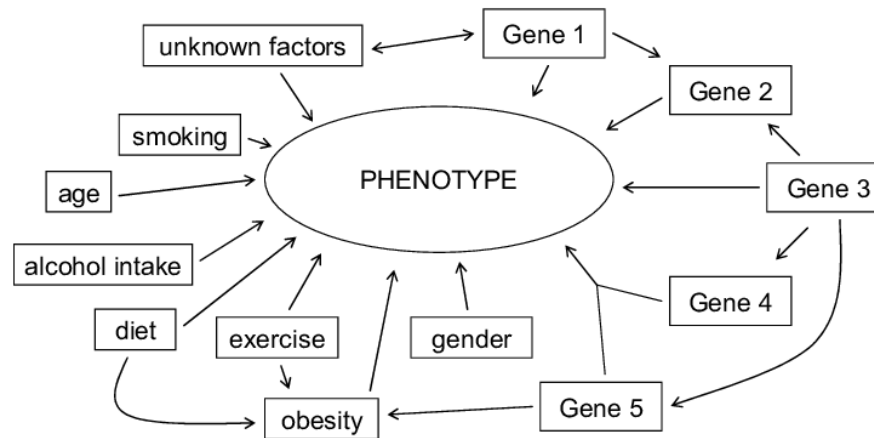
Phenotypic variance of retrieving frequency



The “Multifactorial Model” for complex trait variation

ENVIRONMENTAL FACTORS

GENETIC FACTORS

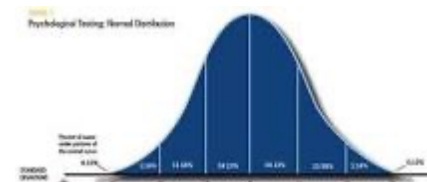
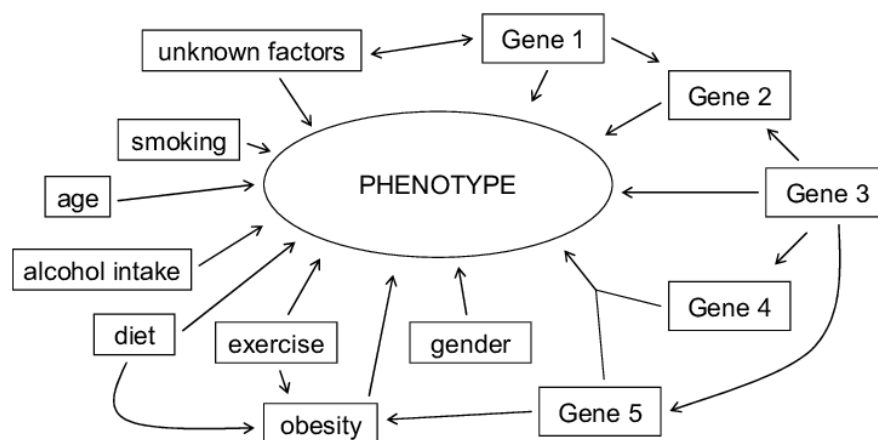


e.g. blood pressure

The phenotypic variance can be decomposed in variance that is attributable to environmental factors and genetic factors

ENVIRONMENTAL FACTORS

GENETIC FACTORS



blood pressure

Main points—Complex (quantitative) traits

- Complex traits can be continuous or show threshold characteristics.
- We are often interested in how much variance there is in the population for the complex trait (the “phenotypic variance”).
- Complex trait variance is typically influenced by genetic variation and environmental variation.
(we will also be interested in “genetic variance” and “environmental variance”)